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1. What is the NLG (Natural Language Generation)?

Natural Language Generation is a part of AI and generates natural language texts from structured data to produce an output. It can be seen as NLP's reverse process, where NLP is used to understand and interpret the natural language to form data, and NLU is used to generate outputs in natural language from structured data.

2. What is the order of steps in natural language understanding?

The order of steps that are to be followed in Natural Language Understanding is as follows:

1. Signal Processing
2. Syntactic Analysis
3. Semantic Analysis
4. Pragmatics

3. What is signal processing in NLP?

Signal processing is a method that enables software to analyze, modify, and synthesize signals. In NLP, these can be sound or text signals.

4. What is pragmatic analysis in NLP?

The pragmatic analysis is the process of information extraction from the given text. It is a set of linguistic and logical tools that enable us to churn out the meaning of the given structure of a text.

5. What is syntactic analysis in NLP?

The syntactic analysis, also referred to as parsing and syntax analysis, is the phase in which we try to process the given text's structure. This process tries to draw meaning from the text by comparing it to formal grammar rules or syntax.

6. What is semantic analysis in NLP?

The semantic analysis is the process of understanding the meaning of the text in the way humans perceive and communicate. It focuses on larger parts of data for processing, as compared to other analysis techniques.

7. What is sentiment analysis in NLP?

The sentiment analysis, also known as opinion mining and emotion AI, is a process of detecting the polarity of the opinion in the text or can be a part of it. It is majorly used to identify, extract, and quantify user or customer reviews' polarity, survey responses, or social media opinions.

8. What is discourse analysis in NLP?

Discourse is a structured group of the sentence. Discourse analysis can be termed as an approach to analyzing the discourse, i.e., text or language. It involves text interpretations and interactions.

9. What is pragmatic ambiguity in NLP?

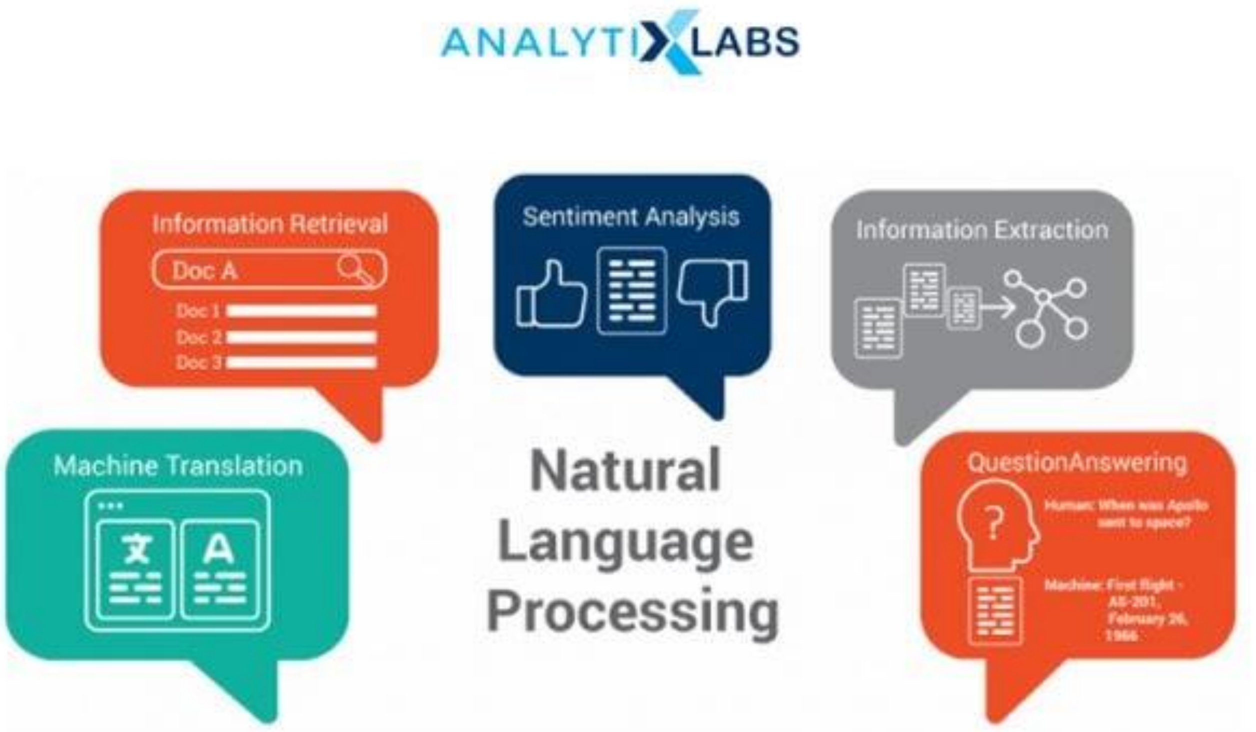
Pragmatic ambiguity can be referred to as a condition where words have multiple interpretations. This condition arises when the meaning of words is not specific; i.e., it can give different meanings.

10. What are the major applications of NLP?

The major applications of NLP are:

1. Machine Translation
2. Speech Recognition

3. Sentiment Analysis
4. Text Classification



11. List any real-world application of NLP?

The most used real-world application of NLP is speech recognition. Examples of speech recognition applications are Amazon Alexa, Google Assistant, Siri, HP Cortana.

12. What are the common NLP techniques?

The common NLP techniques for text extraction are:

1. Named Entity Recognition
2. Sentiment Analysis
3. Text Summarization
4. Aspect Mining
5. Text Modelling

What are the components of NLP?

The components of NLP are:

1. Lexical Analysis
2. Syntactic Analysis
3. Semantic Analysis
4. Discourse Integration
5. Pragmatic Analysis

14. What are the tools used for training NLP models?

The tools used to train NLP models are NLTK, spaCY, PyTorch-NLP, openNLP.

15. Which NLP technique uses a lexical knowledge base to obtain the correct base form of the words?

The **NLP techniques** that use lexical knowledge to obtain the correct base form are lemmatization and stemming.

16. List the models to reduce the dimensionality of data in NLP.

The commonly used models are TF-IDF, Word2vec/Glove, LSI, Topic Modelling, Elmo Embeddings.

17. List some open-source libraries for NLP.

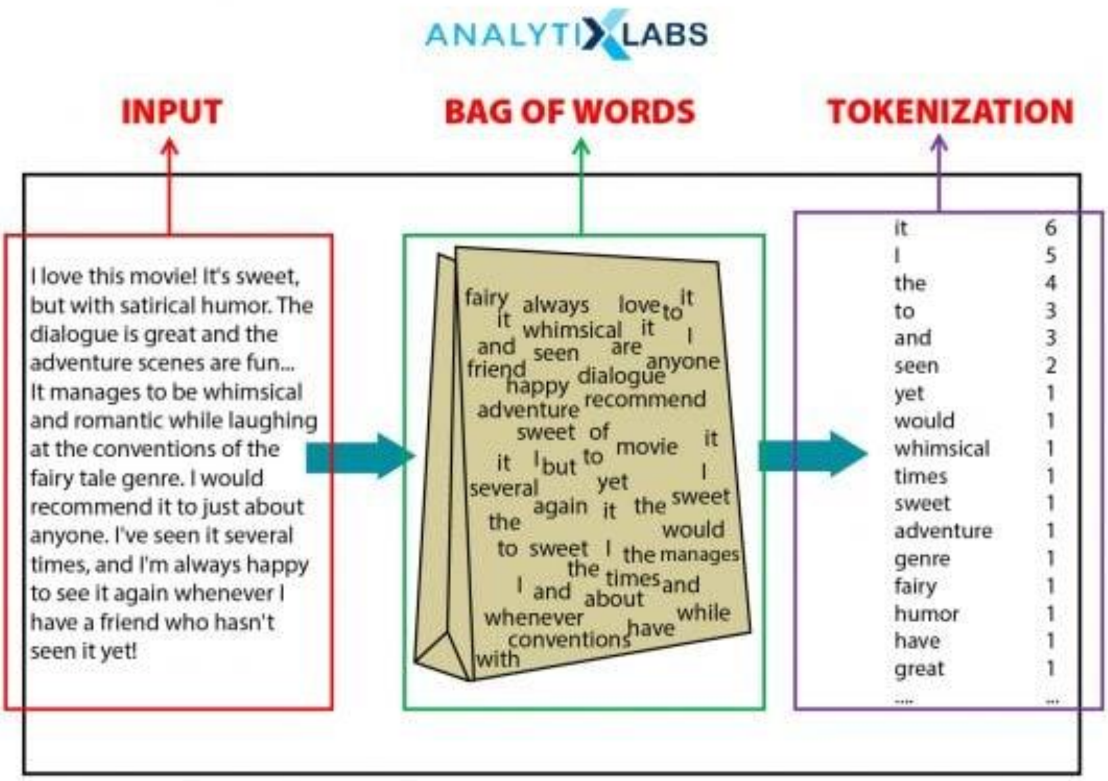
The popular libraries are NLTK (Natural Language ToolKit), SciKit Learn, Textblob, CoreNLP, spaCY, Gensim.

18. Explain the masked language model.

Masked modeling is an example of autoencoding language modeling. Here the output is predicted from corrupted input. By this model, we can predict the word from other words present in the sentences.

19. What is the bag of words model?

The Bagofwords model is used for information retrieval. Here the text is represented as a multiset, i.e., a bag of words. We don’t consider grammar and word order, but we surely maintain the multiplicity.



20. What is CBOW in NLP?

CBOW or continuous bag of words is a model that tries to predict the target word from the available source context words, i.e., the surrounding words. Here the context words are taken into account as multiple words for a given target word.

21. What is TF-IDF and what are its uses?

TF-IDF is an abbreviation for the term frequency-inverse documentary frequency. It is used to provide a numeric value to a word to show how important it is in the document or a corpus.

22. What are POS and tagging?

Parts Of Speech (POS) are the functions of the word, like a noun, verb, etc., and tagging is labeling the words present in the sentences into different parts of speech.

23. What is n-gram in NLP?

N-grams are the continuous sequence (similar to the power set in number theory) of n-tokens of a given text.

24. What is skip-gram?

Skip gram is an unsupervised learning technique used to find the most related words to a target word. It is a reverse process of the continuous bag of words model.

25. What is the corpus in NLP?

Corpus or corpora (plural), is a collection of the text of a similar type, for example, movie reviews, social media posts, etc.

26. What are the features of the text corpus in NLP?

The features of text corpus are:

1. Word count
2. Vector notation
3. Part of speech tag
4. Boolean feature
5. Dependency grammar

27. What is normalization in NLP?

Normalization is the process of mapping similar terms to a canonical form, i.e., a single entity.

28. What is keyword normalization?

Keyword normalization is an **NLP technique** where we apply normalization on a word to condense it to its most basic form.

29. What is lemmatization in NLP?

Lemmatization is a type of normalization used to group similar terms to their base form-based on the parts of speech. For example, talking and talking can be mapped to a single term, walk.

30. What is stemming in NLP?

Stemming in NLP is also a type of normalization and is similar to lemmatization, but the difference here is that it segregates the words without the parts of speech tags. It is faster than lemmatization and can also be more accurate in some cases.

31. What is ambiguity in NLP?

Ambiguity can be referred to as a condition when a word can have multiple interpretations and results in being misunderstood. Natural languages are ambiguous and can make it difficult to process **NLP techniques** on them, resulting in the wrong output.

32. What is tokenization in NLP?

Tokenization is the process of breaking down large sets of text into small parts for easy readability and understanding. Each small part is referred to as 'text' and provides a piece of meaningful information.

33. What are stop words in NLP?

Stop words are the unwanted text that is present in the input. It is the process of removal of unwanted text from further processing of text, for example, a, to, can, etc.

34. How to find word similarity in NLP?

Word similarity in NLP is done by calculating the word vectors of the words in the vector space and then calculating the similarity on a scale of 0 to 1.

35. How to find sentence similarity in NLP?

Sentence similarity is done in NLP by finding the cosine similarity between the two sentences. It can be done by finding the cosine angle between the vectors of two sentences in the inner product space.

36. How to find document similarity in NLP?

Document similarity is done in NLP by converting the documents into the TF-IDF vectors form and finding their cosine similarity.

37. What are transformers?

Transformers are deep learning architectures that can parallelize computations. They are used to learn long-term dependencies.

38. What are punctuations in NLP, and how can we remove them?

Punctuations are the punctuations in the corpus or the input text. We can remove them by using the tokenizer function of NLTK. We can use nltk.RegexpTokenizer() to remove all punctuations.

39. What is latent semantic indexing (LSI)?

Latent Semantic Indexing,, also referred to as latent semantic analysis, is an **NLP technique** used to remove stop words from processing the text into the text's main content. It is used to find relationships between different words.

40. What is named entity recognition (NER)?

Named Entity Recognition is a part of information retrieval, a method to locate and classify the entities present in the unstructured data provided and convert them into predefined categories.

41. What is NLTK in NLP?

NTLK, an abbreviation of Natural Language Toolkit, is one of NLP's most popular libraries. It was written in Python and contained libraries for tokenization, classification, tagging, stemming, parsing, and semantic reasoning.

42. What is spaCY?

spaCY is an open-source library for natural language processing on an advanced level. It is mostly used for production-level usage and uses convolutional neural network models.

43. What is openNLP?

openNLP is a java based library used for Natural Language Processing, and it supports most of the NLP tasks such as tokenization, language detection, etc.

44. What is the difference between NLTK and openNLP?

There is a small difference between NTLK and openNLP, i.e., NLTK is written in python, and openNLP is based on java. One other difference is that NTLK has an option of downloading corpora by an in-built method.

45. What is parsing?

Parsing is the method of analyzing the sentence automatically based on the syntactic structure.

46. What is dependency parsing?

Dependency parsing, also called syntactic parsing, recognizes a dependency parse of a sentence and assigns a syntax structure to a sentence. It focuses on the relationship between different words.

47. What is semantic parsing?

Semantic parsing is a method of conversion of natural language into machine-understandable form.

48. What is constituency parsing?

Constituency parsing is a method of division of sentences into sub-parts or constituencies. It aims to extract a constituency-based parse tree from the constituencies of the sentences.

49. What is shallow parsing?

Shallow parsing, also known as light parsing and chunking, identifies constituents of sentences and then links them to different groups of grammatical meanings.

50. What are the differences between dependency parsing and shallow parsing?

The difference between shallow parsing and dependency parsing is that shallow parsing is the parsing of limited parts of the information. In contrast, dependency parsing is the process of finding relations between all the different words.

51. What is language modeling?

Language modeling is the process of creating a probability distribution of a sequence of words. It is used to provide probability to all the words present in the sequence.

52. What is topic modeling?

Topic modeling is a method of finding abstract topics in a document or set of documents to find hidden semantic structures.

53. What is text summarization in NLP?

Text summarization in NLP is the process of conversion of large pieces of text to short text. It is intended to summarize the given text, keeping the main contents and overall meaning intact.

54. What is the difference between a regular expression and regular grammar?

The difference between regular and regular grammar is that regular grammar is used to generate regular language, and regular expression is used to represent regular language.

55. What is perplexity in NLP?

Perplexity is the condition when the system encounters something unaccountable or which is not meaningful.

56. What is the Naive Bayes algorithm, and where is it used in NLP?

Naive Bayes algorithm is used to predict tags of text by calculating the probability for each tag for the text and then providing the one with the highest probability.

57. What is the PageRank algorithm?

Google uses the PageRank algorithm. It is the algorithm to rank web pages in the search engine results.

58. What is noise removal?

Noise removal is one of the **NLP techniques** i.e., used to remove pieces of text from the corpus that is not necessary as they can hinder our text analysis.

59. What is word embedding?

Word embedding is the process of mapping words from the vocabulary to vectors of real numbers.

60. What are the word embedding libraries?

The libraries that provide word embedding features are spaCY and genism.

61. What is word2vec?

Word2vec is a collection of models that are used to generate word embeddings. These models are trained to reconstruct the linguistic context of the words in the corpus.

61. What is doc2vec?

Doc2vec is one of the unsupervised algorithms used to generate vectors of sentences or documents irrespective of their length.

63. What is a document-term matrix?

The document-term matrix, also called the term-document matrix, is the matrix that describes the frequency of terms occurring in a document.

64. What is wordnet?

Wordnet can be described as a database created to store words from different languages connected by their semantic relationships.

65. What is GloVe in NLP?

The gloVebased on their pronunciation.

66. What is a flexible string matching?

Flexible string matching or fuzzy string matching is a method to find strings that are likely to match a specific pattern. It is also called approximate string matching as it uses an approximation to find patterns between strings.

67. What is cosine similarity?

Cosine similarity is the measure of cosine difference between two non-zero vectors in the inner product space. It is used to find the similarity between documents irrespective of their size.

68. What is information extraction?

Information extraction is the process of extracting useful data in a structured way from a given unstructured set of data.

69. What is object standardization, and when is it used?

Object standardization is the process of extracting useful information from abbreviations and other acronyms that can not be meaningful in lexical dictionaries.

71. What is text generation, and when is it done?

Text-generation is the process of generating natural language texts automatically in response to the communication. It uses artificial intelligence and computational linguistic knowledge to perform this task.

71. How can we estimate the entropy of the English language?

N-grams can estimate the entropy of the English language. The entropy of a letter is calculated by knowing the entropy of all the previous letters.

72. What is Latent Dirichlet Allocation?

Latent Dirichlet Allocation is a topic modeling method where each topic represents a set of words, and every document is made of various words.

73. What are the conditional random fields?

Conditional Random Fields (CRFs) are a collection of statistical modeling methods. It is used for pattern recognition and structure prediction.

74. What are the hidden Markov random fields?

Hidden Markov Random fields are a derivation of the Hidden Markov Model. It is a process generated by a Markov chain, whose state sequence can only be observed by a sequence of observations.

75. What is a coreference resolution?

Coreference resolution is the process of collecting all the expressions that are referring to the same entity in a text. It is used in information extraction, document summarization, and question answering.

76. What is PAC learning?

Probably Approximately Correct learning is a mathematical analysis framework. It is used for the analysis of generalization error of the learning algorithms.

77. What is sequence learning?

Sequence learning is a method of learning where both input and output are sequences.

78. What is an ensemble method?

The ensemble method uses multiple learning algorithms to get enhanced and more accurate performance compared to the performance of an algorithm alone.

So these were the most frequently asked **NLP interview questions**, prepare them well, and increase your chances of getting selected.

SET-2

1. What is Naive Bayes algorithm, When we can use this algorithm in NLP?

Naive Bayes algorithm is a collection of classifiers which works on the principles of the Bayes' theorem. This series of NLP model forms a family of algorithms that can be used for a wide range of classification tasks including sentiment prediction, filtering of spam, classifying documents and more.

Naive Bayes algorithm converges faster and requires less training data. Compared to other discriminative models like logistic regression, Naive Bayes model it takes lesser time to train. This algorithm is perfect for use while working with multiple classes and text classification where the data is dynamic and changes frequently.

2. Explain Dependency Parsing in NLP?

Dependency Parsing, also known as Syntactic parsing in NLP is a process of assigning syntactic structure to a sentence and identifying its dependency parses. This process is crucial to understand the correlations between the "head" words in the syntactic structure.
The process of dependency parsing can be a little complex considering how any sentence can have more than one dependency parses. Multiple parse trees are known as ambiguities. Dependency parsing needs to resolve these ambiguities in order to effectively assign a syntactic structure to a sentence.

Dependency parsing can be used in the semantic analysis of a sentence apart from the syntactic structuring.

3. What is text Summarization?

Text summarization is the process of shortening a long piece of text with its meaning and effect intact. Text summarization intends to create a summary of any given piece of text and outlines the main points of the document. This technique has improved in recent times and is capable of summarizing volumes of text successfully.

Text summarization has proved to a blessing since machines can summarise large volumes of text in no time which would otherwise be really time-consuming. There are two types of text summarization:

- Extraction-based summarization

- Abstraction-based summarization

4. What is NLTK? How is it different from Spacy?

NLTK or Natural Language Toolkit is a series of libraries and programs that are used for symbolic and statistical natural language processing. This toolkit contains some of the most powerful libraries that can work on different ML techniques to break down and understand human language. NLTK is used for Lemmatization, Punctuation, Character count, Tokenization, and Stemming. The difference between NLTK and Spacey are as follows:

- While NLTK has a collection of programs to choose from, Spacey contains only the best-suited algorithm for a problem in its toolkit
- NLTK supports a wider range of languages compared to Spacey (Spacey supports only 7 languages)
- While Spacey has an object-oriented library, NLTK has a string processing library
- Spacey can support word vectors while NLTK cannot

5. What is information extraction?

Information extraction in the context of Natural Language Processing refers to the technique of extracting structured information automatically from unstructured sources to ascribe meaning to it. This can include extracting information regarding attributes of entities, relationship between different entities and more. The various models of information extraction includes:

- Tagger Module
- Relation Extraction Module
- Fact Extraction Module
- Entity Extraction Module
- Sentiment Analysis Module
- Network Graph Module
- Document Classification & Language Modeling Module

6. What is Bag of Words?

Bag of Words is a commonly used model that depends on word frequencies or occurrences to train a classifier. This model creates an occurrence matrix for documents or sentences irrespective of its grammatical structure or word order.

7. What is Pragmatic Ambiguity in NLP?

Pragmatic ambiguity refers to those words which have more than one meaning and their use in any sentence can depend entirely on the context. Pragmatic ambiguity can result in multiple interpretations of the same sentence. More often than not, we come across sentences which have words with multiple meanings, making the sentence open to interpretation. This multiple interpretation causes ambiguity and is known as Pragmatic ambiguity in NLP.

8. What is Masked Language Model?

Masked language models help learners to understand deep representations in downstream tasks by taking an output from the corrupt input. This model is often used to predict the words to be used in a sentence.

9. What is the difference between NLP and CI(Conversational Interface)?

The difference between NLP and CI is as follows:

Natural Language Processing (NLP)	Conversational Interface (CI)
NLP attempts to help machines understand and learn how language concepts work.	CI focuses only on providing users with an interface to interact with.
NLP uses AI technology to identify, understand, and interpret the requests of users through language.	CI uses voice, chat, videos, images, and more such conversational aid to create the user interface.

10. What are the best NLP Tools?

Some of the best NLP tools from open sources are:

- SpaCy
- TextBlob
- Textacy
- Natural language Toolkit (NLTK)
- Retext
- NLP.js
- Stanford NLP
- CogcompNLP

11. What is POS tagging?

Parts of speech tagging better known as POS tagging refer to the process of identifying specific words in a document and grouping them as part of speech, based on its context. POS tagging is also known as grammatical tagging since it involves understanding grammatical structures and identifying the respective component.

POS tagging is a complicated process since the same word can be different parts of speech depending on the context. The same general process used for word mapping is quite ineffective for POS tagging because of the same reason.

12. What is NES?

Name entity recognition is more commonly known as NER is the process of identifying specific entities in a text document that are more informative and have a unique context. These often denote places, people, organizations, and more. Even though it seems like these entities are proper nouns, the NER process is far from identifying just the nouns. In fact, NER involves entity chunking or extraction wherein entities are segmented to categorize them under different predefined classes. This step further helps in extracting information.